ELEC2302/9302 Signals and Systems — Quiz 2 formula sheet

Fourier series (u T-periodic)

Exponential form:

$$\phi_k(t) = e^{jk\omega_0 t}, \qquad c_k = \langle u, \phi_k \rangle_P = \frac{1}{T}\int_0^T u(t)\, e^{-jk\omega_0 t}\, \mathrm{d}t, \qquad u(t) = \sum_{k=-\infty}^{\infty} c_k\, e^{jk\omega_0 t}, \qquad c_0 = \frac{1}{T}\int_0^T u(t)\, \mathrm{d}t.$$

Sine–cosine form (u real): $u(t) = a_0 + \sum_{k=1}^{\infty} \big(a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)\big)$, where

$$a_0 = c_0, \quad a_k = 2\,\mathrm{Re}(c_k) = \frac{2}{T}\int_0^T u(t)\cos(k\omega_0 t)\, \mathrm{d}t, \quad b_k = -2\,\mathrm{Im}(c_k) = \frac{2}{T}\int_0^T u(t)\sin(k\omega_0 t)\, \mathrm{d}t, \quad c_k = \tfrac{1}{2}\left(a_k - jb_k\right).$$

u real	$c_{-k} = \overline{c_k}$	linearity	$\alpha u_1 + \beta u_2$ has coefficients $\alpha c_{1,k} + \beta c_{2,k}$
u even	$b_k = 0$	time shift	$u(t - t_0)$ has coefficients $e^{-jk\omega_0 t_0}\, c_k$
u odd	$a_0 = a_k = 0$	Parseval	$P(u) = \frac{1}{T}\int_0^T \lvert u(t)\rvert^2\, \mathrm{d}t = \sum_{k=-\infty}^{\infty} \lvert c_k\rvert^2$

Fourier series of standard waveforms

$\mathrm{sinc}(x) = \dfrac{\sin(\pi x)}{\pi x}$.

Signal	**Coefficients**
$\cos(\omega_0 t)$	$c_1 = c_{-1} = \frac{1}{2}$, all others zero
$\sin(\omega_0 t)$	$c_1 = -\frac{j}{2}$, $\quad c_{-1} = \frac{j}{2}$, all others zero
Square wave, even: $+A$ for $\lvert t\rvert < T/4$, $-A$ for $T/4 < \lvert t\rvert < T/2$	$c_k = A\,\mathrm{sinc}\left(\frac{k}{2}\right)$ (k odd), $\quad c_k = 0$ (k even)
Pulse train: $A\,p_\tau(t)$ repeated with period T, duty cycle $d = \tau/T$	$c_k = A\,d\,\mathrm{sinc}(kd)$
Triangle wave: $A\,(1 - 4\lvert t\rvert/T)$ for $\lvert t\rvert \le T/2$	$c_k = \frac{4A}{\pi^2 k^2}$ (k odd), $\quad c_k = 0$ (k even)
Sawtooth: $2At/T$ for $\lvert t\rvert < T/2$	$c_0 = 0$, $\quad c_k = \frac{jA(-1)^k}{\pi k}$
Rectified sine: $A\,\lvert\sin(\omega_m t)\rvert$, period $T = \pi/\omega_m$, so $\omega_0 = 2\omega_m$	$c_k = \frac{2A}{\pi\,(1 - 4k^2)}$

Fourier transform

$$U(j\omega) = \int_{-\infty}^{\infty} u(t)\, e^{-j\omega t}\, \mathrm{d}t, \qquad u(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} U(j\omega)\, e^{j\omega t}\, \mathrm{d}\omega, \qquad u \text{ periodic: } U(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} c_k\, \delta(\omega - k\omega_0).$$

$u(t)$	$U(j\omega)$	$u(t)$	$U(j\omega)$
$\delta(t)$	1	$p_\tau(t)$	$\tau\,\mathrm{sinc}\left(\frac{\omega\tau}{2\pi}\right)$
$\delta(t - t_0)$	$e^{-j\omega t_0}$	$\frac{\sin(Wt)}{\pi t}$	$p_{2W}(\omega)$, that is 1 for $\lvert\omega\rvert \le W$ and 0 otherwise
1	$2\pi\delta(\omega)$	$1 - \frac{\lvert t\rvert}{\tau}$ for $\lvert t\rvert < \tau$	$\tau\,\mathrm{sinc}^2\left(\frac{\omega\tau}{2\pi}\right)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$	$e^{-at}H(t)$, $a > 0$	$\frac{1}{a + j\omega}$
$\cos(\omega_0 t)$	$\pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$	$t\,e^{-at}H(t)$	$\frac{1}{(a + j\omega)^2}$
$\sin(\omega_0 t)$	$-j\pi\delta(\omega - \omega_0) + j\pi\delta(\omega + \omega_0)$	$e^{-a\lvert t\rvert}$	$\frac{2a}{a^2 + \omega^2}$
$H(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$	$e^{-at}\cos(\omega_0 t)H(t)$	$\frac{a + j\omega}{(a + j\omega)^2 + \omega_0^2}$
$\mathrm{sgn}(t)$	$\frac{2}{j\omega}$	$e^{-at}\sin(\omega_0 t)H(t)$	$\frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$

Properties of the Fourier transform

	$u(t)$	$U(j\omega)$		$u(t)$	$U(j\omega)$
linearity	$\alpha u_1(t) + \beta u_2(t)$	$\alpha U_1(j\omega) + \beta U_2(j\omega)$	differentiation	$\frac{\mathrm{d}}{\mathrm{d}t} u(t)$	$j\omega\, U(j\omega)$
time shift	$u(t - t_0)$	$e^{-j\omega t_0}\, U(j\omega)$	integration	$\int_{-\infty}^{t} u(\varsigma)\,\mathrm{d}\varsigma$	$\frac{U(j\omega)}{j\omega} + \pi U(0)\delta(\omega)$
frequency shift	$e^{j\omega_0 t} u(t)$	$U\big(j(\omega - \omega_0)\big)$	multiply by t	$t\, u(t)$	$j \frac{\mathrm{d}}{\mathrm{d}\omega} U(j\omega)$
modulation	$u(t)\cos(\omega_0 t)$	$\frac{1}{2} U\big(j(\omega - \omega_0)\big) + \frac{1}{2} U\big(j(\omega + \omega_0)\big)$	convolution	$(h * u)(t)$	$H(j\omega)\, U(j\omega)$
time scaling	$u(at)$	$\frac{1}{\lvert a \rvert} U\left(\frac{j\omega}{a}\right)$	multiplication	$u(t)\, y(t)$	$\frac{1}{2\pi}\,(U * Y)(j\omega)$
time reversal	$u(-t)$	$U(-j\omega)$	duality	$U(jt)$	$2\pi\, u(-\omega)$

Parseval's theorem:

$$\int_{-\infty}^{\infty} u(t)\, \overline{y}(t)\,\mathrm{d}t = \frac{1}{2\pi} \int_{-\infty}^{\infty} U(j\omega)\, \overline{Y}(j\omega)\,\mathrm{d}\omega, \qquad E(u) = \int_{-\infty}^{\infty} \lvert u(t) \rvert^2\,\mathrm{d}t = \frac{1}{2\pi} \int_{-\infty}^{\infty} \lvert U(j\omega) \rvert^2\,\mathrm{d}\omega.$$

Second order systems

$$G(j\omega) = \frac{K\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}, \qquad G(0) = K,$$

where $\omega_n > 0$ is the natural frequency and ζ the damping ratio.

$\zeta > 1$	overdamped	distinct real poles
$\zeta = 1$	critically damped	repeated pole $-\omega_n$
$\zeta < 1$	underdamped	$-\zeta\omega_n \pm j\omega_d$, $\quad \omega_d = \omega_n\sqrt{1-\zeta^2}$ the damped natural frequency

$$\omega_r = \omega_n\sqrt{1 - 2\zeta^2}, \qquad \lvert G(j\omega_r) \rvert = \frac{\lvert K \rvert}{2\zeta\sqrt{1-\zeta^2}}, \qquad \zeta < \frac{1}{\sqrt{2}}.$$

Stability

Exponentially stable: there exist $M > 0$ and $a > 0$ such that $\lvert h(t) \rvert \le Me^{-at}$ for all $t \ge 0$.
Marginally stable: there exists $M > 0$ such that $\lvert h(t) \rvert \le M$ for all $t \ge 0$, and $h(t) \not\to 0$ as $t \to \infty$.
BIBO stable: $\sup_{t \ge 0} \lvert u(t) \rvert < \infty$ implies $\sup_{t \ge 0} \lvert y(t) \rvert < \infty$ for every input u.